Лабораторная работа

Номер 7

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Информация

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Цель работы

Получить навыки настройки межсетевого экрана в Linux в части переадресации портов и настройки Masquerading.

Запуск сервера

```
C:\work\nsandryushin\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Resuming suspended VM...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
==> server: Machine booted and ready!
==> server: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> server: flag to force provisioning. Provisioners marked to run always will still run.
==> server: Running provisioner: common hostname (shell)...
```

Рис. 1: Запуск сервера

Копирование конфигурационного файла

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/se
rvices/ssh-custom.xml
[root@server.nsandryushin.net ~]# cd /etc/firewalld/services/
[root@server.nsandryushin.net services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on
 remote machines. It provides secure encrypted communications. If you plan on accessing yo
ur machine remotely via SSH over a firewalled interface, enable this option. You need the
openssh-server package installed for this option to be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
```

Рис. 2: Копирование конфигурационного файла

Редактирование файла /etc/firewalld/services/ssh-custom.xml

```
  GNU nano 8.1                 /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH-New</short>
  <description>Long SSH description</description>
  <port protocol="tcp" port="2022"/>
</service>
```

Рис. 3: Редактирование файла /etc/firewalld/services/ssh-custom.xml

Список служб firewalld

```
[root@server.nsandryushin.net services]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp am
qps anno-1602 anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-dire
ctor bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-
iv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-uni
cast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker-registry docke
r-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galer
a ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 https id
ent imap imaps iperf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins
kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube
-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-s
ecure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kube
let-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-c
lient llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mongodb
 mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wa
nted netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ov
irt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3
 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv
 ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind
 rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-col
lection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap
 spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer steam-streaming
stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing synct
hing-gui syncthing-relay synergy syscomlan syslog syslog-tls telnet tentacle terraria tftp
 tile38 tinc tor-socks transmission-client turn turns upnp-client vdsm vnc-server vrrp war
pinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host
ws-discovery-tcp ws-discovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbi
x-web-service zero-k zerotier
```

Рис. 4: Список служб firewalld

Списки служб

```
[root@server.nsandryushin.net services]# firewall-cmd --reload
success
[root@server.nsandryushin.net services]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp am
qps anno-1602 anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-dire
ctor bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-
iv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-uni
cast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker-registry docke
r-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galer
a ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 https id
ent imap imaps iperf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins
kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube
-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-s
ecure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kube
let-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-c
lient llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mongodb
 mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wa
nted netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ov
irt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3
 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv
 ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind
 rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-col
lection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap
 spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-transfer steam
-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn sync
thing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-tls telnet tentacle te
rraria tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsm vnc-serv
er vrrp warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disc
overy-host ws-discovery-tcp ws-discovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh x
mpp-client xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-tr
apper zabbix-web-service zero-k zerotier
[root@server.nsandryushin.net services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh
```

Рис. 5: Списки служб

Добавление службы как активной

```
[root@server.nsandryushin.net services]# firewall-cmd --add-service=ssh-custom
success
[root@server.nsandryushin.net services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh ssh-custom
[root@server.nsandryushin.net services]# firewall-cmd --add-service=ssh-custom --permanent
success
[root@server.nsandryushin.net services]# firewall-cmd --reload
success
```

Рис. 6: Добавление службы как активной

Форвардинг портов

```
[root@server.nsandryushin.net services]# firewall-cmd --add-forward-port=port=2022:proto=t
cp:toport=22
success
```

Рис. 7: Форвардинг портов

Подключение по ssh

```
[nsandryushin@client.nsandryushin.net ~]$ ssh -p 2022 nsandryushin@server.nsandryushin.net
The authenticity of host '[server.nsandryushin.net]:2022 ([192.168.1.1]:2022)' can't be es
tablished.
ED25519 key fingerprint is SHA256:1PtjeoTNJyZ4YiNPkJxzegCAqkN7GCRWznJJBakANJY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server.nsandryushin.net]:2022' (ED25519) to the list of known
 hosts.
nsandryushin@server.nsandryushin.net's password:
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last login: Thu Oct 16 17:05:55 2025
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 8: Подключение по ssh

```
[root@server.nsandryushin.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server.nsandryushin.net services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90
-forward.conf
[root@server.nsandryushin.net services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server.nsandryushin.net services]# firewall-cmd --zone=public --add-masquerade --per
manent
success
[root@server.nsandryushin.net services]# firewall-cmd --reload
success
```

Рис. 9: Включение перенаправления и маскарадинга

Проверка доступа в интернет

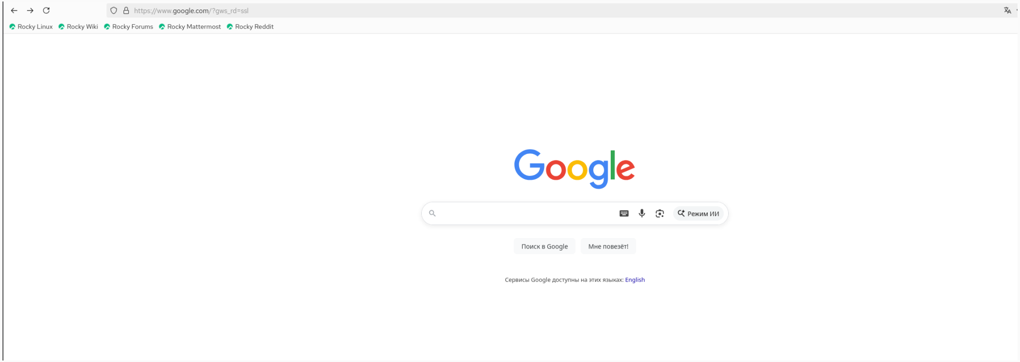

Рис. 10: Проверка доступа в интернет

Сохранение конфигурации vagrant

```
[root@server.nsandryushin.net services]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# mkdir -p /vagrant/provision/server/firewall/etc/fir
ewalld/services
[root@server.nsandryushin.net server]# mkdir -p /vagrant/provision/server/firewall/etc/sys
ctl.d
[root@server.nsandryushin.net server]# cp -r /etc/firewalld/services/ssh-custom.xml /vagra
nt/provision/server/firewall/etc/firewalld/services/
[root@server.nsandryushin.net server]# cp -r /etc/sysctl.d/90-forward.conf /vagrant/provis
ion/server/firewall/etc/sysctl.d/
[root@server.nsandryushin.net server]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# touch firewall.sh
[root@server.nsandryushin.net server]# chmod +x firewall.sh
[root@server.nsandryushin.net server]# nano firewall.sh
[root@server.nsandryushin.net server]#
```

Рис. 11: Сохранение конфигурации vagrant

firewall.sh

```
  GNU nano 8.1                                       firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рис. 12: firewall.sh

```
    server.vm.provision "server http",
                        type: "shell",
                        preserve_order: true,
                        path: "provision/server/http.sh"
    server.vm.provision "server mysql",
                        type: "shell",
                        preserve_order: true,
                        path: "provision/server/mysql.sh"
    server.vm.provision "server firewall",
                        type: "shell",
                        preserve_order: true,
                        path: "provision/server/firewall.sh"

```

Рис. 13: Vagrantfile

Выводы

В результате выполнения лабораторной работы были получены навыки работы с фаерволом и настройкой форвардинга и маскарадинга